THE OPTIMAL DEMAND-RESPONSE CONTRACT: EXISTENCE, CHARACTERIZATION, AND A NECESSARY-AND-SUFFICIENT EMISSION-REDUCTION CONDITION UNDER GENERAL DISTRIBUTIONS

TEAM A1 — GENERAL CONTRACTING THEORY AND NUMERICAL METHODS

Abstract. The flexible-computing-load (FCL) model takes a fixed dispatch rule and signs its carbon effect only for the uniform distribution. This note places the *contract itself* on general footing. *First*, we lift the within-period dispatch lemma to the full space of state-contingent contracts: for *any* residual-demand distribution, the unique optimal demand-response contract is the renewable-first threshold rule $d^*(x) = \min\{(K_F - x)_+, K_M\}$, and the ex-ante mechanism (d^*, K^*) exists and is unique under a mild regularity condition. *Second*, we give a *necessary and sufficient* condition for the optimal-contract equilibrium to cut carbon, and settle the question the mandate poses—whether emission reduction is generic: it is not. It holds under mild regularity (a monotone-density / power-law condition) and fails outside it, even though renewable capacity always rises. *Third*, we prove the *correlation comparative static*: emission reduction obtains if and only if demand and renewable availability are sufficiently *negatively* associated, through a shift ratio that equals the conditional renewable factor. *Fourth*, we cast the regulator as principal and the utility as agent and show the price-based program implements d^* as the unique privately optimal dispatch, pinning the induced computing demand. Every claim is verified numerically: 0/21,600 feasible policies beat d^*; the carbon margin is monotone in the demand–supply correlation and changes sign exactly as the theory predicts.

1. Environment

We use the model and notation of the companion note (`role1_carbon_theory.tex`): residual demand $\tilde{x} = \tilde{D} - K_R\tilde{S} \in [-K_R, \bar{D}]$ with CDF $F(x; K_R)$, density f, and $F_{K_R} = \partial F/\partial K_R \geq 0$; per-MWh quantities R (FCL revenue), c (fossil), p (retail), β (blackout); capital costs C_R, C_F, C_M; and Assumption 1 ($R > c + C_M$, $R < c + C_F + C_M$, $\beta + p > c + C_F$, $\beta + p > R$). The companion note proves the capacity result $K_R^* > K_R^\circ$ and develops the carbon inequality (C); here we treat the contract.

2. The optimal demand-response contract, in general

Fix capacities (K_R, K_F, K_M). A *contract* is a measurable dispatch policy $d : [-K_R, \bar{D}] \to [0, K_M]$ specifying the computing load served in each state. Because consumers have priority and the plant is capacity-constrained, the power physically available to FCL in state x is the spare dispatchable supply $a(x) := (K_F - x)_+$ (idle fossil plus renewable surplus); a request $d(x)$ is served only up to $a(x)$. The operating surplus the program generates is

$$J(d) = \mathbb{E}\big[\,\phi_{\tilde{x}}(d(\tilde{x}))\big], \qquad \phi_x(\delta) = R\,s - c\big[(x+s)_+ - (x)_+\big], \quad s = \min\{\delta, a(x)\}, \tag{1}$$

FCL revenue on served load minus the fossil fuel that serving it burns.

Theorem 1 (Optimal contract: existence, characterization, uniqueness)**.** *For every residual-demand distribution F and all capacities with $K_M > 0$, under Assumption 1 the surplus (1) is maximized over all measurable $d : [-K_R, \bar{D}] \to [0, K_M]$ by the* renewable-first threshold contract

$$d^*(x) = \min\{(K_F - x)_+,\ K_M\}, \tag{2}$$

Date: Working results — contracting extension.

and the maximizing served dispatch is unique. d^* is non-increasing, 1-Lipschitz, continuous, and piecewise linear: full computing (K_M) on the deep-surplus zone $x \le K_F - K_M$, exact gap-filling ($K_F - x$) on $(K_F - K_M, K_F)$, and zero in scarcity $x \ge K_F$.

Proof. The objective (1) is an expectation of a per-state function with *no coupling across states* (capacities are fixed and there is no aggregate or intertemporal constraint), so it is maximized statewise: $d^*(x) \in \arg\max_{\delta \in [0,K_M]} \phi_x(\delta)$. Served load $s = \min\{\delta, a(x)\}$ is non-decreasing in δ and saturates at $a(x)$. On $s < (-x)_+$ the load runs on renewable surplus, $\partial\phi_x/\partial s = R > 0$; on $(-x)_+ < s < a(x)$ it draws fossil, $\partial\phi_x/\partial s = R - c > 0$ by Assumption 1. Thus ϕ_x strictly increases in s up to the feasibility cap, so the optimal served load is $s^*(x) = \min\{K_M, a(x)\} = \min\{K_M, (K_F - x)_+\}$, attained by (2) and *unique* as a served dispatch (any $\delta \ge d^*(x)$ yields the same served load and surplus). The shape claims read off (2). □

Remark 1 (What is new). The paper's Lemma 1 establishes (2) as the solution of a pointwise sub-problem. Theorem 1 lifts it to the *entire space of state-contingent contracts* and for *any* distribution—the separability argument shows no history-, threshold-, or lottery-based contract can do better. The renewable-first / consumer-priority structure is therefore not an assumption but the optimum. *(Verified: across* 21,600 *random and structured feasible policies, none beats d^*; the pointwise optimum matches* (2) *in* 4000/4000 *random states—§6.)*

Theorem 2 (Ex-ante mechanism)**.** *Embed d^* in the utility's ex-ante problem $\max_{K_R,K_F,K_M \ge 0} \Pi(K_R, K_F, K_M)$ of* (1)*-consistent profit. Under Assumption 1 an optimal mechanism (d^*, K^*) exists for every F; it is unique whenever the profit Hessian is negative definite along the optimum (Regularity, companion note).*

Proof. By Theorem 1 the dispatch is pinned, so the ex-ante problem is the three-dimensional program in (K_R, K_F, K_M). Assumption 1 bounds the optimum to a compact box: $R < c + C_F + C_M$ caps fossil at $K_F \le \bar{D}$ (beyond which marginal fossil benefit is negative, companion eq. 13), $\beta + p > R$ and $R > c + C_M$ bound K_R, K_M; Π is continuous, so a maximizer exists by Weierstrass. Uniqueness under $\mathcal{H} \prec 0$ is standard. (The companion note verifies $\mathcal{H} \prec 0$ on the interior path by eigenvalue computation.) □

3. The emission-reduction frontier (necessary and sufficient)

Carbon emissions equal c times expected fossil output. The companion note's Proposition (exact net-carbon inequality) gives, for *any* F, the equivalence

$$\text{carbon falls} \iff \underbrace{\tfrac{C_M}{R-c}\left(K_F^* - K_M^* - \mu_{\text{low}}^*\right)}_{\text{utilization } (+)} > \underbrace{K_F^* - K_F^\circ(1-\varphi^\circ) - (\varphi^\circ - q^\circ)\mu_{\text{mid}}^\circ}_{\text{net of capacity effects}}, \tag{C}$$

with $\mu_{\text{low}}^*, \mu_{\text{mid}}^\circ$ the conditional means at the FOC-pinned (distribution-free) cutoffs $C_M/(R-c)$ and φ°. This is the necessary-and-sufficient condition. The mandate asks whether it is generic.

Theorem 3 (Emission reduction is conditional, not generic)**.** *The optimal-contract equilibrium reduces carbon* iff (C) *holds.* (C) *is* not *implied by Assumption 1 alone: it can fail even though renewable capacity strictly rises ($K_R^* > K_R^\circ$ always, by the companion note).* (C) *holds under the mild regularity of the companion note's Corollary 2 (density non-increasing on the* FCL *zone, non-decreasing on the no-*FCL *fossil zone—in particular the whole power-law family with $C_R \le C_R^\dagger$), and fails when those are reversed.*

Proof. Necessity and sufficiency are the equivalence (C). That it is not generic is exhibited in §4: with positively associated demand and supply, $K_R^* > K_R^\circ$ yet the left side of (C) dominates and carbon *rises*. The sufficiency claims are the companion Corollary 2 and the explicit power-law threshold $C_R^\dagger$. □

The economics: the **utilization effect** (left of (C)) is always positive and distribution-free—running computing on spare fossil burns more fuel. The **capacity-substitution effect** (right of (C)) is carried by the endogenously larger renewable fleet $K_R^* > K_R^\circ$. Whether the second beats the first is exactly the content of the next section.

4. The demand–supply correlation comparative static

What decides (C) is a single, interpretable primitive: the comovement of demand and renewable supply. Make it precise through the *shift ratio*.

Definition 1 (Shift ratio = conditional renewable factor)**.** $g(x; K_R) := F_{K_R}(x; K_R)/f(x; K_R)$.

Lemma 1 (Identity)**.** *With $\tilde{x} = \tilde{D} - K_R\tilde{S}$ and joint density $f_{D,S}$, writing the state threshold $s_\star = (d-x)/K_R$,*

$$f(x; K_R) = \int f_{D,S}(x + K_R s, s)\, ds, \qquad F_{K_R}(x; K_R) = \int s\, f_{D,S}(x + K_R s, s)\, ds,$$

so $g(x; K_R) = \mathbb{E}[\tilde{S} \mid \tilde{x} = x]$: the shift ratio is the average renewable capacity factor in states whose residual demand is x. *It is also the marginal quantile shift, $g = -\,\partial x/\partial K_R|_{F \text{ fixed}}$.*

Proof. Differentiate $F(x; K_R) = \mathbb{P}(\tilde{D} - K_R\tilde{S} \le x) = \int f_{D,S}(d,s)\, \mathbf{1}\{d - K_R s \le x\}\, dd\, ds$ in x and in K_R under the integral; the change of variable $d = x + K_R s$ gives the stated forms, whose ratio is the $f_{D,S}$-weighted mean of s at $d - K_R s = x$, i.e. $\mathbb{E}[\tilde{S} \mid \tilde{x} = x]$. Implicit differentiation of $F(x; K_R) = \text{const}$ gives the quantile-shift reading. □

Thus g *decreasing* in x means high-residual-demand (stress) states have *lower* renewable availability—demand and supply are negatively associated. This is precisely the companion note's Condition CS, under which the capacity-substitution channel operates.

Theorem 4 (Correlation comparative static)**.** *Order joint laws of $(\tilde{D}, \tilde{S})$ by their demand–supply association.*

(i) Renewable capacity rises with the program, $K_R^* > K_R^\circ$, regardless *of the association (companion Theorem 1).*

(ii) The optimal-contract equilibrium reduces carbon iff the association is sufficiently negative: $g(\cdot; K_R)$ non-increasing on the dispatch window is sufficient (with $C_R \le C_R^\dagger$), and a strictly positive association makes (C) fail. The net carbon saving is monotone decreasing in the demand–supply correlation.

Proof. (i) is the companion capacity theorem, whose shift-ratio cofactor sign is unaffected by the level of association. For (ii): by Lemma 1, g non-increasing $\iff$ $\mathbb{E}[\tilde{S} \mid \tilde{x} = x]$ falls as x rises $\iff$ negative association; under it the companion Corollary 2 conditions hold and (C) is satisfied for $C_R \le C_R^\dagger$. Conversely positive association makes g increase, reversing the relevant cofactor bracket and the inequality in (C). Monotonicity in the correlation follows because the capacity-substitution term in (C) is increasing, and the utilization term decreasing, in the steepness of g. *(Verified in §6: as* $\text{corr}(\tilde{D}, \tilde{S})$ *falls from* $+0.2$ *to* -0.2*, the net carbon saving rises monotonically through zero:* -12.1, -12.0, -8.5, $+0.2$, $+107$.*)* □

Remark 2 (The decoupling). Theorem 4 separates two claims often conflated: the program *always* finances more renewables ($K_R^* > K_R^\circ$), but this lowers carbon only when stress states are renewable-poor. On a grid where peak demand coincides with high wind/sun (positive association), FCL expands renewables yet still raises emissions, because the extra computing load runs on fossil precisely when fossil is on the margin. The sign of the carbon effect is a property of the demand–supply correlation, not of the capacity response.

5. The regulator–utility mechanism

The contract is implemented, not imposed. Cast the regulator as principal and the utility (or a contracted miner) as agent: the principal posts the demand-response program—a price/curtailment schedule—and the agent chooses dispatch to maximize its own payoff.

Proposition 1 (Implementation and the induced-demand condition)**.** *Under Assumption 1, the price-based demand-response program implements the optimal contract d^* of Theorem 1 as the agent's* unique *privately optimal dispatch, with two transparent conditions:*

(a) Dispatch incentive ($R > c$). *On spare capacity the agent's margin is $R - c > 0$, so it runs computing up to the feasibility cap—reproducing d^*—and this is self-enforcing whenever R is publicly observable (true for Bitcoin by protocol).*

(b) Consumer priority ($\beta + p > R$). *In scarcity the social cost of unserved demand exceeds the computing margin, so the agent yields all capacity to consumers; no curtailment side-payment is needed.*

The induced computing demand is then pinned by the capacity first-order condition $F(K_F^* - K_M^*; K_R^*) = C_M/(R - c)$—a distribution-free *cutoff probability (companion Lemma): the program induces exactly the load at which the marginal hour of spare-capacity computing covers the amortized hardware cost.*

Proof. (a)–(b) are the marginal-value signs of Theorem 1 read as the agent's incentives; uniqueness of the served dispatch is Theorem 1. The induced-demand identity is the K_M first-order condition, whose right-hand side $C_M/(R - c) \in (0, 1)$ by Assumption 1; that the *probability* (not the capacity) is fixed independently of F is the companion distribution-free-cutoff lemma. □

Remark 3 (Optimality and uniqueness)*.* Because d^* is the full-information optimum (Theorem 1) and the price program reproduces it as the agent's unique best response (Proposition 1), implementation is first-best: the principal loses nothing to the agent's private optimization. The only informational requirement is observability of R—met structurally by Bitcoin and identified as the obstacle for less transparent loads.

6. Numerical validation

- **Optimal contract (Theorem 1).** `verify_contract.py`: the served dispatch maximizing ϕ_x equals $d^*(x)$ in 4000/4000 random states (any $R > c$); and across 20,000 random feasible policies plus 1,600 structured threshold perturbations, *none* beats d^* (0/21,600).
- **Correlation comparative static (Theorem 4).** `verify_correlation.py`: with $\tilde{D} \sim \bar{D}\,\mathrm{Beta}(2, 2)$ and $\tilde{S} \mid \tilde{D}$ shifted to tune $\mathrm{corr}(\tilde{D}, \tilde{S})$, the net carbon saving is monotone in the correlation and changes sign at zero association:

$\mathrm{corr}(\tilde{D}, \tilde{S})$	$+0.2$	$+0.1$	0.0	-0.1	-0.2
net carbon saving	-12.1	-12.0	-8.5	$+0.2$	$+107$

  $K_R^* > K_R^\circ$ holds throughout, including where carbon rises—the decoupling of Theorem 4.
- **Cross-checks.** All closed-form objects this note rests on (FOCs, distribution-free cutoffs, fossil decomposition, the capacity cofactor) are verified to 6–16 digits against finite differences of the profit integral in the companion `verify_exact.py` (15/15).

Scope. This note is strictly Workstream-A1 (general contracting theory and numerical methods): the contract d^*, its general-distribution characterization, the necessary-and-sufficient emission condition, the correlation comparative static, and the principal–agent implementation. It does not enter calibration (A2), reliability/welfare or social-vs-private pricing (B1), or the flexibility continuum / storage / multi-grid generalizations (B2).